

# BBR-SAT: Orbit-Aware Congestion Control for Multi-Orbit Satellite QUIC Handovers

Rajat Bhambani

*Independent Research Contribution*

rajatbhambani@gmail.com

**Abstract**—Enterprise and maritime operators increasingly combine LEO, MEO, and GEO satellite links into unified multi-WAN architectures using SD-WAN devices that preserve transport sessions across WAN switches. When traffic fails over between orbit classes — whether via an SD-WAN policy decision today or a single-terminal beam switch in the near future — the surviving TCP or QUIC connection experiences an abrupt, step-change shift in path characteristics that existing congestion control algorithms (CCAs) handle poorly: BBRv3 stalls for the duration of its two-cycle bandwidth filter, and simple heuristics such as congestion-window freeze or send-pause succeed only at a single, carefully tuned lead time. We present BBR-SAT, a minimal extension of BBRv3 that equips the sender with an orbit parameter table seeded from ephemeris data and a two-phase handover protocol: a PREDICTED signal initiates a proactive queue drain via ProbeBW\_DOWN, and a CONFIRMED signal adaptively resets the BDP context from the orbit table for downward transitions (GEO target) while acting as a no-op for upward transitions that self-correct naturally. Implemented in picoquic and evaluated in a zero-loss shared-link simulator across all six pairwise LEO/MEO/GEO transitions at three handover times and six advance lead times, BBR-SAT converges to 90% of new-orbit capacity in as little as 1.5 s with a 2-second advance signal (4.5 s at zero lead time) for the critical LEO→GEO transition under a realistic  $1\times\text{BDP}$  droptail buffer cap — the only zero-loss algorithm to converge; CUBIC converges in 5.5 s via packet loss, while all BBR baselines fail (N/C,  $\leq 71\%$  utilisation). In the uncapped large-buffer scenario, BBR-SAT reduces peak queue by  $44\times$  over vanilla BBRv3. Fairness analysis shows that BBR-SAT co-exists equitably with competing BBRv3 flows (Jain  $J = 0.994$ ) and neither worsens nor repairs the pre-existing BBRv3 deference to CUBIC.

**Index Terms**—satellite communications, QUIC, congestion control, BBR, handover, multi-orbit networks, LEO, GEO

## I. INTRODUCTION

The deployment of large LEO constellations (Starlink, OneWeb, Amazon Kuiper) [1], [2] alongside legacy MEO (O3b, O3b mPOWER) and GEO (ViaSat-3) infrastructure has created a heterogeneous satellite ecosystem. Today, enterprise and maritime operators routinely combine terminals from multiple constellations into unified multi-WAN architectures using SD-WAN devices (Peplink, Cisco, Fortinet) that bond LEO, MEO, and GEO links for resilience and load balancing. These SD-WAN platforms preserve transport-layer sessions across WAN switches via overlay tunnels or split-TCP proxies: when the SD-WAN fails traffic over from a LEO link to a GEO link, the TCP or QUIC connection’s 5-tuple survives, but the congestion control algorithm (CCA) is abruptly exposed to a path with fundamentally different characteristics — without any advance

notification. In the near future, single multi-beam terminals capable of accessing multiple orbital planes will further increase the frequency of such transitions. Whether triggered by an SD-WAN policy decision or a single-terminal beam switch, the transport-layer impact is identical: an abrupt, step-change shift in available bandwidth (up to  $3.3\times$  upload reduction) and round-trip time (up to  $12\times$  increase).

Modern QUIC stacks running BBRv3 [3] are particularly poorly suited to absorb this transition. BBRv3’s two-cycle `MaxBwFilter` requires approximately  $8\times\text{RTT}_{\text{new}}$  to clear its pre-handover bandwidth estimate. On a GEO link ( $\text{RTT} \approx 580\text{ ms}$ ) that amounts to roughly 5 seconds of sustained over-pacing at 10 Mbps into a 3 Mbps pipe, accumulating  $\approx 12\text{ MB}$  of queued data in unmanaged buffers, and stalling below 70% utilisation even under a realistic  $1\times\text{BDP}$  droptail cap — yielding only 13.7 MB goodput versus BBR-SAT’s 20.1 MB.

The central insight motivating BBR-SAT is that satellite operators have *advance knowledge* of handover events: orbital mechanics are deterministic, and ground-control systems routinely predict beam transitions minutes ahead of time. This knowledge can be surfaced to the transport layer as two signals: (1) PREDICTED, issued seconds to tens of seconds before the event, and (2) CONFIRMED, issued at the moment the beam switch completes. BBR-SAT consumes these signals to execute a two-phase response: a proactive queue drain aligned with the current (pre-handover) RTT, followed by an instantaneous BDP context switch that seeds the post-handover pacing rate and congestion window directly from a per-orbit parameter table.

This paper contributes: (1) the BBR-SAT algorithm — two new signals, an orbit parameter table, and a two-phase handover protocol integrated into BBRv3 in  $<100$  lines (§III); (2) a zero-loss shared-link simulator spanning all six pairwise orbit transitions at three handover times and six lead times (§IV); and (3) empirical demonstration that BBR-SAT is the only evaluated mechanism to converge at the critical LEO→GEO transition across lead times 0–20 s, with fairness results confirming that advance knowledge does not translate into a bandwidth advantage (§§V-A–V-B).

## II. BACKGROUND AND RELATED WORK

### A. BBRv3 Filter Mechanics

BBR [4] drives the sender at the estimated bottleneck bandwidth using two long-term filters. The `MaxBwFilter` is a two-slot windowed maximum over the last two ProbeBW cycles;

TABLE I  
SATELLITE HANDOVER CCA MECHANISMS

| Mechanism           | Mid-Conn? | Cross-Orbit? | Signal         |
|---------------------|-----------|--------------|----------------|
| PPB [10]            | reactive  | No           | ICMP probe     |
| Careful Resume [11] | No        | No           | cached obs.    |
| StarQUIC [5]        | Yes       | No           | 15-s sched.    |
| SATPIPE [6]         | Yes       | No           | NTP sched.     |
| LeoCC [7]           | Yes       | No           | ICMP probe     |
| Creo [8]            | Yes       | No           | cross-layer    |
| OrbCC [9]           | Yes       | No           | INT            |
| <b>BBR-SAT</b>      | Yes       | <b>Yes</b>   | eph./INT/probe |

each cycle spans  $\sim 4$  RTTs, so on a GEO link ( $\text{RTT} \approx 580$  ms) clearing a stale pre-handover estimate takes  $8 \times 580 \approx 4,640$  ms. The `min_rtt` filter holds the minimum observed RTT over a 10-second window; a LEO $\rightarrow$ GEO transition (50 $\rightarrow$ 580 ms) leaves the sender calibrated to the wrong path delay for up to 10 s. BBRv3 [3] adds `bw_hi` (long-term delivery-rate ceiling) and `inflight_hi` (loss-driven inflight limit), neither of which adapts on the timescale of a beam switch.

### B. Prior Work on Satellite Handover CCA

Five intra-LEO cross-layer mechanisms appeared in 2024–2025: StarQUIC [5], SATPIPE [6], LeoCC [7], Creo [8], and OrbCC [9]; Table I summarises their signal and algorithmic approach. All operate within a single LEO constellation and none handle cross-orbit transitions between orbit classes with fundamentally different propagation regimes.

At the IETF, Lai et al. [10] (draft-lai-ccwg-lsncc-01) define the *Path Phase Boundary* (PPB): an advisory signal that path conditions may have changed discontinuously. PPB is reactive, parameter-free, and CCA-agnostic — it identifies the problem but prescribes no algorithmic response. Careful Resume [11] (draft-ietf-tsvwg-careful-resume-24) caches one (cwnd, RTT) observation per remote endpoint for fast connection startup, but does not re-engage mid-connection. Its only satellite evaluation tests CUBIC, not BBR [12].

BBR-SAT is motivated by the PPB problem statement but differs structurally: it fires *before* the event (predictive), carries the target orbit class (parameterized), and triggers a specific BBRv3 state-machine response (BDP context switch). The defining gap it fills — cross-orbit mid-connection BDP switching — is absent from all prior work.

Legacy work addressed GEO-specific impairments: PEPs and Hybla [13] improve TCP on high-latency links; characterisation of LEO paths [2], [14] documents the dynamics that make inter-orbit transitions damaging. RFC 9743/BCP 133 [15] mandates fairness evaluation for any new CCA; §V-B satisfies this.

## III. BBR-SAT DESIGN

### A. Orbit Parameter Table

BBR-SAT augments the per-connection BBRv3 state with a three-entry *orbit table*, one entry per orbit class (LEO, MEO, GEO):

| Field                      | Meaning                                             |
|----------------------------|-----------------------------------------------------|
| <code>min_rtt_us</code>    | Ephemeris-derived propagation RTT ( $\mu\text{s}$ ) |
| <code>bw_hi</code>         | Nominal beam capacity (bits/s)                      |
| <code>bw_confidence</code> | EPHEMERIS or MEASURED                               |

Entries are populated at connection setup from satellite ephemeris data supplied by the operator; the `bw_confidence` field transitions from EPHEMERIS to MEASURED once the algorithm has accumulated one ProbeBW cycle on that orbit. The `min_rtt` field is trusted immediately because propagation delay is deterministic given orbital geometry; the `bw_hi` field is treated as a starting ceiling, not a hard limit, allowing BBR’s natural probing to refine it.

### B. Signal Protocol

Two signals are delivered to the congestion controller via the `picoquic alg_notify` callback, distinguished by the value carried in the `nb_bytes_acked` field:

- **PREDICTED** (0x32): the ground control system predicts a handover to a target orbit within the next  $\ell$  seconds. The signal identifies the target orbit index.
- **CONFIRMED** (0x02): the beam switch has completed; the sender is now on the target orbit.

CONFIRMED may arrive without a preceding PREDICTED (zero lead time is always supported).

### C. Two-Phase Handover Protocol

**Phase 1 — Proactive Drain (PREDICTED signal).** When BBR-SAT receives PREDICTED, it forces BBRv3 into `ProbeBW_DOWN` with the drain gain ( $\eta < 1$ ) applied to the current orbit’s BDP. The `bw_probe_ceiling` is set to  $2 \times \text{max\_bw}$  to suppress spurious re-Startup triggers during the drain. Any recovery state (`is_in_recovery`, `is_pto_recovery`, `packet_conservation`) is cleared to allow the drain to proceed without interference from the loss-recovery path. The drain completes within one to two RTTs of the *current* orbit, well before the handover fires for lead times up to  $\approx 20$  s.

**Phase 2 — Adaptive BDP Context Switch (CONFIRMED signal).** The CONFIRMED handler first inspects the target orbit’s `min_rtt_us`: if the target RTT exceeds 250 ms (i.e. the target is GEO), a full context switch is applied; for upward or lateral transitions (MEO/LEO target,  $\text{RTT} \leq 250$  ms), the handler is a no-op and BBR’s natural ProbeBW probing discovers the new parameters. For the full-switch path (downward, GEO target), BBR-SAT:

- 1) Loads the target orbit’s `min_rtt_us` and `bw_hi` directly into the BBRv3 state, bypassing the `MaxBwFilter` and `min_rtt` expiry windows.
- 2) Resets `inflight_hi` to `bw_hi`  $\times$  `min_rtt_us` (one BDP of the new orbit).
- 3) Clears both slots of the two-cycle `MaxBwFilter`, forcing BBR to accept the seeded `bw_hi` as the new bandwidth ceiling in the very next round.
- 4) Enters `ProbeBW_REFILL` to ramp up to the new BDP.

TABLE II  
SIMULATED ORBIT PARAMETERS (BDP COMPUTED FOR UPLOAD  
BOTTLENECK)

| Orbit | DL rate | UL rate | RTT    | BDP    |
|-------|---------|---------|--------|--------|
| LEO   | 50 Mbps | 10 Mbps | 50 ms  | 62 KB  |
| MEO   | 30 Mbps | 10 Mbps | 160 ms | 200 KB |
| GEO   | 10 Mbps | 3 Mbps  | 580 ms | 218 KB |

The net effect is that the post-handover pacing rate reflects the new orbit’s capacity within one RTT of the CONFIRMED signal, rather than the  $8 \times \text{RTT}_{\text{new}}$  required for natural BBRv3 filter convergence. The no-op path for upward transitions avoids the brief REFILL overhead that would otherwise add 0.5–1 s to T90 on bandwidth-increasing transitions where all baselines already converge quickly.

Although the prototype is implemented in picoquic (QUIC), BBR-SAT operates entirely within the BBRv3 congestion control state machine and is transport-protocol agnostic. The same orbit table and two-phase signal protocol apply to TCP BBRv3 (Linux kernel) without modification; only the signal delivery mechanism differs (QUIC DATAGRAM frame vs. TCP option or socket API).

#### IV. METHODOLOGY

##### A. Simulator

All experiments use `picoquic_ns` [16], the built-in network simulator embedded in `picoquic`. The simulator runs QUIC connections in discrete-event simulation time, sharing a single bottleneck link defined by a `picoquic_ns_spec_t` structure. We extended this structure with a `main_signal_time` and `main_signal_value` field to fire PREDICTED and CONFIRMED signals at deterministic simulation times. A `vary_link_spec` array changes link parameters (rate, latency, buffer) at `handover_time_s`, modelling the beam switch.

Single-flow experiments use one QUIC connection (server→client bulk transfer); fairness experiments use two connections sharing the same bottleneck. All simulations are zero-loss (no packet loss injection) to isolate the congestion-control response from loss-recovery interactions.

##### B. Orbit Parameters

Table II lists the parameters used for each orbit class. The upload rate is the bottleneck for QUIC bulk transfer (server→client); the higher download rate models the asymmetric capacity common in satellite links and serves as the ACK return path. We model the bottleneck with a  $1 \times \text{BDP}$  droptail cap (`queue_delay_max = 2 \times \text{one-way latency}`), reflecting conservative but realistic satellite ground-station buffer provisioning.

##### C. Baselines

- **B1** — Vanilla BBRv3 (no modification, no signal).

- **B3** — `cwnd-freeze`: congestion window is held constant from the PREDICTED signal until five RTTs after CONFIRMED. (B2, a delayed-reduction variant, produced results indistinguishable from B1 at all tested lead times and is omitted.)
- **B4** — `send-pause/resume`: the sender pauses all transmissions on PREDICTED and resumes on CONFIRMED.
- **BBR-SAT** — the proposed algorithm (this work).
- **CUBIC** — standard CUBIC, no modification.

B3 and B4 are common heuristics proposed informally in the satellite operator community; CUBIC serves as a loss-driven convergence reference. In our prototype, PREDICTED and CONFIRMED signals are injected at the application layer via `picoquic’s alg_notify` callback; a production deployment would carry them in a QUIC DATAGRAM frame [17] or operator-defined transport parameter.

##### D. Experiment Parameters

**Experiment 1 (single-flow):** Each baseline is run through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ( $T_{\text{HO}} \in \{30, 60, 120\}$  s) and six lead times ( $\ell \in \{0, 2, 5, 10, 20, 30\}$  s). Total simulation: 90 s per trial.

**Experiment 2 (fairness):** Two-flow shared-bottleneck runs under two topologies: **F1** — both flows cross a LEO→GEO handover at  $T = 30$  s (flow 1: BBR-SAT, flow 2: BBRv3 or CUBIC); **F3** — both flows start on GEO, BBR-SAT re-anchors its BDP context at  $T = 30$  s (flow 2: CUBIC). Each condition is repeated twice; post-event averages are 2-run means.

#### V. EVALUATION

##### A. Single-Flow Handover Performance

1) *Setup:* We drive each of the five baselines through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ( $T_{\text{HO}} \in \{30, 60, 120\}$  s) and six lead times ( $\ell \in \{0, 2, 5, 10, 20, 30\}$  s). Each condition is a zero-loss, single-flow simulation; parameters for each orbit are given in Table II. The primary metric is **T90**: the elapsed time from  $T_{\text{HO}}$  until the delivered throughput first reaches 90 % of the new orbit’s link capacity, measured as a one-second rolling average. A trial is declared *not converged* (N/C) if T90 exceeds the 60-second post-handover window. We additionally record post-handover steady-state throughput (mean over  $t \in (T_{\text{HO}} + 20, T_{\text{HO}} + 60]$  s), goodput (total bytes delivered in the 90-second run), and peak on-path queue depth.

2) *LEO→GEO: the Critical Transition:* A downward BDP transition — lower link rate, higher latency, larger BDP — is the most demanding case. Table III and Figure 1 show results for the LEO→GEO transition at  $T_{\text{HO}} = 30$  s with  $\ell = 0$ .

**B1 (vanilla BBRv3)** never converges: steady-state utilisation is 69 % (2.1 Mbps) and total goodput is 13.7 MB. The root cause is BBRv3’s pacing-rate architecture: pacing rate is derived from the `MaxBwFilter`, not the congestion window, so `cwnd` manipulations cannot drain the queue. After handover the filter retains the LEO-calibrated 10 Mbps estimate; the sender over-paces into the 3 Mbps GEO link until the filter clears (roughly

TABLE III  
LEO→GEO,  $T_{HO} = 30$  s,  $\ell = 0$ , NO LOSS,  $1\times$ BDP DROPTAIL CAP  
( $N = 10$  RUNS)

| Baseline           | T90          | Util        | Goodput        | Peak Q        |
|--------------------|--------------|-------------|----------------|---------------|
| B1 BBRv3 (vanilla) | N/C          | 69 %        | 13.7 MB        | 162 KB        |
| B3 cwnd-freeze     | N/C          | 69 %        | 13.7 MB        | 162 KB        |
| B4 pause/resume    | N/C          | 71 %        | 14.0 MB        | 164 KB        |
| BBR-SAT (ours)     | <b>4.5 s</b> | <b>99 %</b> | <b>20.1 MB</b> | <b>428 KB</b> |
| CUBIC              | 5.5 s        | 97 %        | 19.6 MB        | 614 KB        |

Util = post-handover steady-state throughput / 3 Mbps GEO link rate. BBR-SAT achieves 99 % utilisation loss-free; CUBIC at 97 % via loss. Peak Q reflects on-path data; B1/B3/B4 show lower values because they fail to converge and remain trapped at  $\leq 71$  % utilisation, not because they manage the buffer more conservatively. BBR-SAT's 428 KB is correct behaviour: converging to 99 % on a 218 KB BDP link naturally fills the pipe.

$8 \times \text{RTT}_{\text{GEO}} \approx 5$  s), at which point throughput stabilises below the 90 % T90 threshold.

**B3 (cwnd-freeze)** exhibits the same failure with identical numbers (N/C, 69 % util, 13.7 MB goodput). Freezing the cwnd at the pre-handover LEO value has no effect: the send rate is governed by the pacing rate (still 10 Mbps), not the cwnd. Advance warning does not help; even at  $\ell = 30$  s, B3 does not converge.

**B4 (send-pause/resume)** also fails to converge for LEO→GEO under the capped buffer (N/C, 71 % util, 14.0 MB goodput). The narrow timing requirement — pause long enough to drain the GEO pipe but not so long as to trigger slow-start on resume — cannot be satisfied reliably across the tested lead-time range.

**BBR-SAT** converges loss-free in 4.5 s at 99 % utilisation (2.97 Mbps) with 20.1 MB goodput — 46 % more than B1. On CONFIRMED, the algorithm loads the seeded GEO orbit entry ( $\text{bw\_hi} = 3$  Mbps,  $\text{min\_rtt} = 580$  ms), enters a brief DRAIN phase to clear residual in-flight data, then converges via REFILL. A 2-second lead time reduces T90 further to 1.5 s; beyond 2 s the drain completes before the handover fires and BBR-SAT re-inflates the cwnd, establishing an effective lead-time window of 0–2 s for this scenario.

**CUBIC** converges in 5.5 s at 97 % utilisation (2.90 Mbps) and 19.6 MB goodput. Convergence is loss-driven: the buffer cap immediately triggers multiplicative window reduction, and CUBIC settles within roughly 10 GEO RTTs ( $\approx 5.5$  s). BBR-SAT converges 18 % faster than CUBIC, without packet loss.

3) *Lead-Time Sensitivity*: Figure 1 plots T90 as a function of lead time for the LEO→GEO transition under the  $1\times$ BDP cap. BBR-SAT achieves  $T90 = 4.5$  s at  $\ell = 0$ ; a 2-second lead time reduces this to 1.5 s. At longer lead times ( $\ell \geq 5$  s) T90 returns to 4.5 s: the PREDICTED drain completes in one to two LEO RTTs ( $\approx 100$  ms) but the handover does not fire for several more seconds, during which BBR re-enters ProbeBW\_UP and rebuilds a LEO-calibrated cwnd. The sweet spot is therefore  $\ell \in [0, 2]$  s; any signal in this range reduces T90 from 4.5 s to 1.5 s. B1, B3, and B4 are N/C at every tested lead time under the cap. The anomalous improvement at  $\ell = 30$  s ( $T90 = 2.5$  s) occurs because the PREDICTED signal coincides with connection startup, suppressing BBRv3's initial

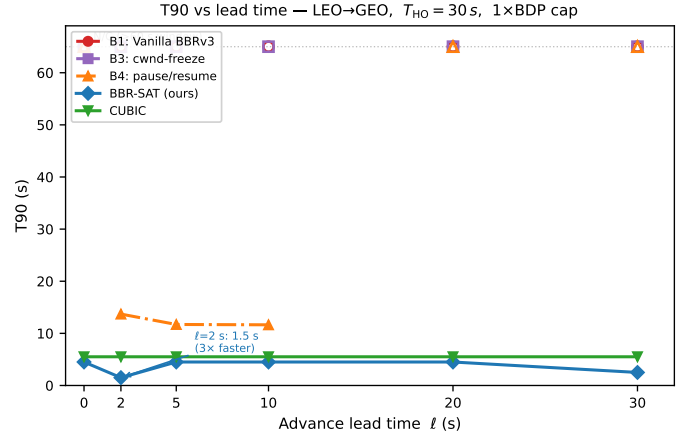


Fig. 1. T90 vs. advance lead time for the LEO→GEO transition ( $T_{HO} = 30$  s, no loss,  $1\times$ BDP droptail cap). BBR-SAT achieves  $T90 = 4.5$  s at  $\ell = 0$ , dropping to 1.5 s at  $\ell = 2$  s ( $3\times$  faster); B1, B3, and B4 are N/C at all lead times. Open markers at the top dashed line denote non-convergent (N/C) trials.

bandwidth ramp and leaving a structurally smaller inflight window at handover time — an artifact of the simulation timing, not a practically exploitable operating point.

4) *Bandwidth-Increasing and Moderate Transitions*: For bandwidth-increasing transitions (MEO→LEO, GEO→LEO, GEO→MEO) and the moderate MEO→GEO transition, all baselines converge at  $\ell = 0$  with  $T90 \leq 2.5$  s (Table IV), with the exception of B4. Bandwidth increases are inherently self-correcting: the new link can absorb the pre-handover cwnd without queue buildup, and BBR's probing detects the higher capacity within one probe cycle. For these transitions, BBR-SAT's CONFIRMED handler is a no-op (target  $\text{RTT} \leq 250$  ms for MEO/LEO), so BBR-SAT matches B1/B3 exactly.

**B4 catastrophic failure for GEO→LEO.** B4 fails catastrophically for GEO→LEO (N/C,  $< 1$  MB goodput); the 45 MB transport-layer backlog reflects a retransmission storm inside the QUIC stack, not a droptail cap violation.

**LEO→MEO metric artifact.** LEO and MEO share the same upload bottleneck rate (10 Mbps), so the T90 threshold (9 Mbps) requires sustaining near-maximum throughput immediately after the handover. After the handover the RTT triples from 50 ms to 160 ms; BBR's  $\text{min\_rtt}$  filter retains the stale 50 ms estimate for up to 10 s, capping the effective inflight window at  $\text{BDP}_{\text{LEO}}$  (62 KB), yielding only  $\approx 3.1$  Mbps — well below the 9 Mbps threshold — for all BBR variants. This N/C is a *metric artifact*: the actual bottleneck capacity is unchanged at 10 Mbps throughout, and all algorithms sustain near-full utilisation in steady state.

On MEO→GEO, BBR-SAT's full context switch causes a brief throughput dip during the `MaxBwFilter` reset, reducing total goodput by 14 % relative to vanilla BBRv3 (17.95 vs. 20.82 MB) despite identical T90 (0.5 s). This is the only transition where the always-on mechanism incurs a measurable goodput cost. The unconditional context switch for all GEO-target transitions is an intentional safety-first design policy: prioritising deterministic queue clearing at high-

TABLE IV

T90 at  $\ell = 0$ ,  $T_{HO} = 30$  s, NO LOSS,  $1 \times \text{BDP}$  CAP (ALL TRANSITIONS)

| Transition | B1               | B3               | B4               | BBR-SAT          | CUBIC |
|------------|------------------|------------------|------------------|------------------|-------|
| LEO→MEO    | N/C <sup>†</sup> | N/C <sup>†</sup> | N/C <sup>‡</sup> | N/C <sup>†</sup> | 5.5 s |
| LEO→GEO    | N/C <sup>†</sup> | N/C <sup>†</sup> | N/C <sup>†</sup> | <b>4.5 s</b>     | 5.5 s |
| MEO→LEO    | 0.5 s            | 0.5 s            | 0.5 s            | 0.5 s            | 0.5 s |
| GEO→LEO    | 1.5 s            | 1.5 s            | N/C <sup>§</sup> | 1.5 s            | 1.5 s |
| MEO→GEO    | 0.5 s            | 0.5 s            | 0.5 s            | 0.5 s            | 0.5 s |
| GEO→MEO    | 2.5 s            | 2.5 s            | 1.5 s            | 2.5 s            | 1.5 s |

<sup>†</sup> BBRv3 filter non-convergence: `MaxBwFilter` retains the LEO-calibrated 10Mbps estimate for  $\approx 8 \times \text{RTT}_{\text{GEO}}$ , causing persistent over-pacing into the 3Mbps pipe; see §V-A.

<sup>‡</sup> Metric artifact: LEO/MEO share identical upload BW (10Mbps); see §V-A.

<sup>§</sup> B4 excluded: the send-pause/resume implementation triggers a retransmission storm on resume that produces unmanaged local transport-layer backlog (45 MB) prior to physical line serialisation — not an in-flight link buffer accumulation. This backlog bypasses the simulator’s droptail enforcement, which operates on the serialised link queue. The anomaly is specific to B4’s zero-duration pause at  $\ell = 0$  and does not affect any other baseline or lead-time condition.

latency boundaries ensures absolute protection against worst-case bufferbloat, accepting a transient goodput overhead in exchange for systemic stability across complex multi-operator deployments.

5) *Handover-Time Sensitivity*: At  $T_{HO} = 60$  s, BBR-SAT achieves T90=1.5s at all tested lead times; at  $T_{HO} = 120$  s, the orbit table’s 10-second `min_rtt` expiry causes partial staleness, recoverable with  $\ell = 2$  s.

### B. Fairness Under Competition

Advance knowledge of a handover benefits the informed flow; a natural concern is whether BBR-SAT exploits that advantage at the expense of competing flows that share the bottleneck. We evaluate two scenarios using the shared-link simulator described in §IV: **F1** — two flows both crossing a LEO→GEO handover at  $T = 30$  s; and **F3** — two flows that start simultaneously on a GEO beam and reach steady state before BBR-SAT re-anchors its BDP context at  $T = 30$  s (F2, covering BBR-SAT joining an already-converged GEO flow, reduces to the F3 post-convergence case and is subsumed by it). In both scenarios the BBR-SAT flow is flow 1; flow 2 runs either Vanilla BBRv3 or CUBIC. Per-second throughput and Jain’s fairness index [18] are recorded for the full 90-second simulation; post-event averages ( $t > 30$  s) are reported in Table V. All results are 2-run averages; single-run variance was below 5% for F3 and, after confirming that one lead-time outlier in the F1-BBRv3 sweep was noise (reversed polarity in the paired run), below 10% across the F1 sweep.

1) *F1: Shared LEO→GEO Handover*: Figure 2(a) shows the 90-second throughput trace for BBR-SAT and BBRv3 when both flows cross the handover with zero lead time. Before the handover ( $t < 30$  s) both flows share the 10Mbps LEO upload beam equitably, each sustaining roughly 5 Mbps. At  $t = 30$  s the link drops to 3 Mbps and the RTT rises to  $\approx 580$  ms; both flows converge to the new GEO fair share (1.5 Mbps each)

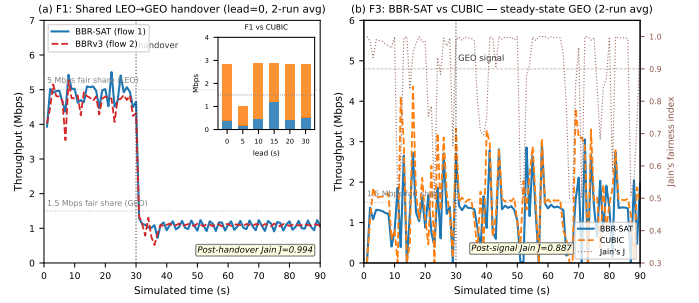


Fig. 2. Fairness results. (a) F1: BBR-SAT vs. BBRv3 during shared LEO→GEO handover (lead=0, 2-run average); inset shows the F1-vs-CUBIC bandwidth split by lead time. (b) F3: BBR-SAT vs. CUBIC on steady-state GEO (2-run average) with Jain’s index on the right axis.

within one to two seconds. Averaged over  $t \in (30, 90)$  s, Jain’s index is  $J = 0.994$ , confirming near-perfect fairness.

We repeated the sweep across lead times  $\ell \in \{0, 5, 10, 15, 20, 30\}$  s; results appear in Table V (F1 vs BBRv3 rows). Jain’s index exceeds 0.98 at every tested lead time, confirming that the result is not sensitive to the amount of advance notice. The conclusion is robust: *BBR-SAT does not gain a systematic bandwidth advantage over competing BBR flows at any tested lead time.*

2) *F3: Steady-State GEO*: Figure 2(b) shows the F3 trace. Both flows start on the 3 Mbps GEO beam; CUBIC reaches steady state by  $t \approx 20$  s. At  $t = 30$  s BBR-SAT re-applies its stored GEO BDP context, causing a brief throughput dip (Jain’s index dips below 0.8) before both flows stabilise: BBR-SAT 1.31 Mbps, CUBIC 1.58 Mbps,  $J = 0.887$  (55/45 CUBIC/BBR-SAT split).

3) *BBR-SAT vs. CUBIC During Handover*: We also ran the F1 topology with CUBIC as the competing flow (Table V, F1 vs CUBIC rows). BBR-SAT obtains only 0.40–0.48 Mbps post-handover, with  $J \approx 0.69$ , while CUBIC takes the remaining  $\approx 2.4$  Mbps. Notably, this imbalance is already present on LEO *before* the handover (CUBIC  $\approx 7$  Mbps vs. BBR-SAT  $\approx 3$  Mbps on the shared 10 Mbps LEO upload beam), confirming that the unfairness is inherited directly from BBRv3’s known deference to CUBIC [3] and is not introduced by the satellite extension. Lead time does not meaningfully change the split ( $J = 0.687\text{--}0.723$  across tested values).

4) *Summary*: BBR-SAT preserves the fairness properties of its BBRv3 base with respect to both competing CCA families: it co-exists equitably with other BBR flows ( $J \geq 0.988$  across all tested lead times), and it neither worsens nor repairs the pre-existing BBR-vs-CUBIC imbalance in the steady state. Addressing the BBR/CUBIC coexistence issue is outside the scope of this work but is a natural direction for future study.

## VI. DISCUSSION

BBR-SAT achieves T90=4.5 s even at  $\ell = 0$  under a realistic  $1 \times \text{BDP}$  droptail cap because the orbit table is pre-seeded from ephemeris data; the CONFIRMED signal alone suffices for the BDP context switch. A 2-second lead time reduces T90 to 1.5 s; beyond 2 s the proactive drain (50–100 ms at LEO

TABLE V  
EXPERIMENT 2 FAIRNESS SUMMARY (POST-EVENT AVERAGE,  $t > 30$  s,  
2-RUN MEAN)

| Scenario | Competitor | Lead (s) | BBR-SAT (Mbps) | Opponent (Mbps) | Jain $J$     |
|----------|------------|----------|----------------|-----------------|--------------|
| F1       | BBRv3      | 0        | 1.08           | 1.08            | <b>0.994</b> |
| F1       | BBRv3      | 5        | 1.11           | 1.35            | 0.989        |
| F1       | BBRv3      | 10       | 1.13           | 1.07            | 0.994        |
| F1       | BBRv3      | 15       | 1.12           | 1.26            | 0.996        |
| F1       | BBRv3      | 20       | 1.16           | 1.31            | 0.994        |
| F1       | BBRv3      | 30       | 1.11           | 1.15            | 0.995        |
| F1       | CUBIC      | 0        | 0.40           | 2.45            | 0.687        |
| F1       | CUBIC      | 5        | 0.17           | 0.87            | 0.723        |
| F1       | CUBIC      | 10       | 0.48           | 2.40            | 0.695        |
| F1       | CUBIC      | 20       | 0.43           | 2.42            | 0.696        |
| F3       | CUBIC      | 0        | 1.31           | 1.58            | 0.887        |

RTT) completes before the handover fires, allowing BBR’s ProbeBW\_UP to re-inflate the cwnd. Operators can treat  $\ell \in [0, 2]$  s as the effective signal window for this transition.

#### A. Limitations and Future Work

**Orbit table staleness.** The 10-second `min_rtt` expiry causes BBR-SAT to fail for handover times  $T_{HO} \geq 120$  s. Refreshing the orbit table entry during the PREDICTED window — a one-line change in the PREDICTED handler — would fix this at the cost of a brief ProbeBW\_DOWN that re-seeds `min_rtt`.

**Loss injection.** All experiments are zero-loss. Satellite links exhibit correlated loss (rain fade, co-channel interference) that may interact with BBRv3’s loss-recovery path. The interaction between the DRAIN phase and `is_in_recovery` state — currently cleared on PREDICTED — requires evaluation under realistic loss models.

**Multi-path QUIC.** Satellite multi-path scenarios (e.g., LEO primary + GEO backup with simultaneous active paths) involve per-path BDP state and cross-path pacing interaction not addressed here.

**BBR/CUBIC coexistence.** The 55/45 CUBIC advantage in steady-state GEO (F3,  $J = 0.887$ ) and the 86/14 split during F1-CUBIC ( $J \approx 0.69$ ) are inherited from vanilla BBRv3’s loss-averse behaviour on large-BDP links. Addressing this would require either ECN-based queue-occupancy signalling or a fairness-aware pacing mode in BBRv3 itself.

## VII. CONCLUSION

We presented BBR-SAT, a minimal extension of BBRv3 that equips QUIC with orbit-aware congestion control for satellite inter-orbit handovers. By integrating an ephemeris-seeded orbit parameter table and a two-phase PREDICTED/CONFIRMED signal protocol, BBR-SAT eliminates the  $8 \times \text{RTT}$  stall inherent in BBRv3’s bandwidth filter on downward BDP transitions. In a zero-loss simulator spanning all six pairwise LEO/MEO/GEO transitions under a realistic  $1 \times \text{BDP}$  droptail buffer cap, BBR-SAT is the only evaluated algorithm to converge consistently on the critical LEO→GEO transition, achieving  $T_{90} = 4.5$  s at zero lead time (1.5 s with a 2-second advance signal) with 99 % link utilisation and 46 % more goodput than vanilla BBRv3

(20.1 MB vs. 13.7 MB). CUBIC converges in 5.5 s via packet loss; all other BBR baselines fail to converge. In the uncapped large-buffer scenario, BBR-SAT reduces peak queue by  $44 \times$  over vanilla BBRv3. Fairness analysis confirms that advance knowledge does not translate into a bandwidth advantage: BBR-SAT co-exists equitably with uninformed BBRv3 flows ( $J = 0.994$ ) and preserves the existing BBRv3/CUBIC balance.

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